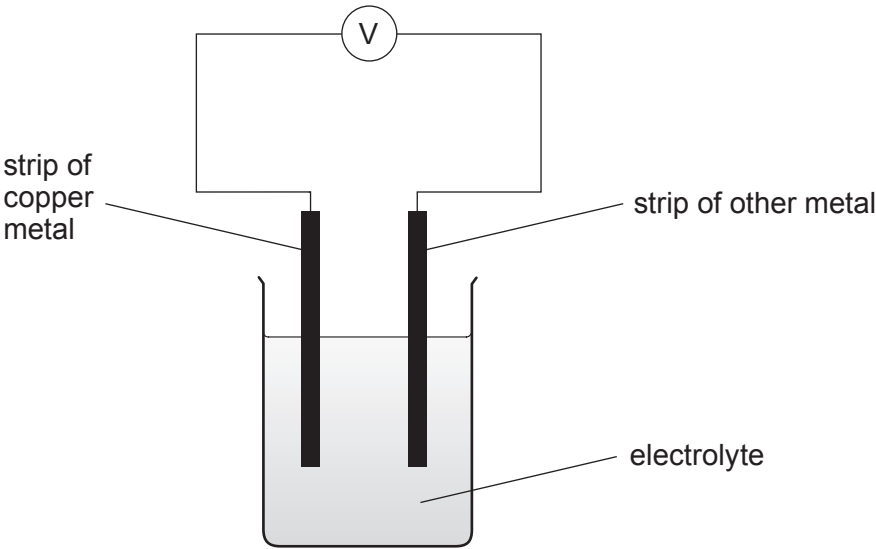


3. When two different metals are connected in a cell, **the metal with the higher reactivity transfers its electrons to the other metal.**

The potential difference produced between pairs of metals can be used to place them in order of reactivity. **The bigger the potential difference, the bigger the difference in reactivity.**

(a) The following apparatus was used to investigate the reactivity of four different metals, **A**, **B**, **C** and **D**, compared with copper.



Each metal was placed separately into a cell with a copper strip. The potential difference was recorded for each metal and the results are shown below.

Metal	Potential difference (V)	Direction of electron flow
A	0.3	copper → metal A
B	0.6	metal B → copper
C	1.1	metal C → copper
D	0.8	copper → metal D

Use the information to place metals **A**, **B**, **C** and **D** in order of their reactivity in relation to copper. [2]

most reactive



1.

2.

3. copper

4.

5.



- (b) The reactivity of four other metals, **W**, **X**, **Y** and **Z**, was also investigated using the same apparatus. Some of the results are shown in the following table.

Pair of metals in the cell	Potential difference (V)	Direction of electron flow
W and X	1.2	W → X
W and Y	0.9	Y → W
W and Z	0.8	
Y and Z		Y → Z

The order of reactivity of these four metals is as follows.

most reactive

Y
Z
W
X

Use this information to give

- (i) the direction of the electron flow when **W** and **Z** are placed in the cell, [1]

.....

- (ii) the potential difference for the cell with metals **Y** and **Z**. [1]

..... V

- (c) When copper and zinc are placed into the cell, the following reaction takes place.



Explain how **this** reaction shows both oxidation and reduction. [2]

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